

Gayana Shaminda Karunaratne

Software Engineer | Full-Stack Engineer | AI/ML

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Profile

Computer Engineering graduate with 10 months of industry experience in software development, contributing to production applications using Next.js, React, TypeScript, and modern web technologies. Hands-on experience across frontend and backend development, REST APIs, databases, automated testing, cloud deployment, application security, and AI-powered systems using Python, Node.js, Django, and FastAPI. Interested in building reliable, scalable, and maintainable software across full-stack and AI-focused engineering environments.

Technical Skills

Frontend: JavaScript, TypeScript, React, Next.js, Tailwind CSS

Backend & APIs: Python, Django, Django REST Framework, FastAPI, Node.js, Express, REST APIs, WebSockets

Databases: PostgreSQL, MongoDB, Redis

Testing & Development Tools: Automated Testing, Git, GitHub, Postman, Jira

Core Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Microservices

Cloud & Deployment: Docker, AWS, Azure, CI/CD, Linux

Additional Technologies: Java, C++, Spring Boot, Jenkins, Terraform, Ansible

Languages: Sinhala (Native), English (Professional)

Experience

Intern Software Engineer

Apr 2025 – Feb 2026

BotCalm (Private) Limited

- Developed frontend modules across four role-based portals (Admin, Teacher, Student, Parent) within a large-scale school management platform structured as a Turborepo monorepo with shared UI and authentication packages.
- Implemented data-heavy interfaces, sortable and filterable tables, multi-step forms with Zod validation, and a rich-text content editor with file upload.
- Built a real-time chat feature from scratch (WebSocket, emoji picker, voice recording, message search) used across all portals.
- Investigated and resolved software defects during a live QA and release cycle, collaborating with developers and QA engineers.
- Contributed to the development of responsive marketing frontends using Next.js, TypeScript, and Tailwind CSS based on Figma designs.

Tools Used: Next.js, React, TypeScript, Tailwind CSS, TanStack Table, React Hook Form, Zod, ShadCN UI, WebSockets, Git, Jira

Projects

AI-Powered Project Evaluator & Viva Examiner [github](#)

Sep 2026

Full-Stack Web Development | Django | PostgreSQL

- Built a full-stack AI viva platform with a Django backend containing 25+ data models and a Next.js frontend with 90+ components, supporting live viva sessions and role-based dashboards.
- Implemented RAG-based question generation with dual-tier LLM routing and sentence-transformer retrieval.
- Integrated Agora real-time video with cloud recording and an asynchronous worker for post-session analysis, using PostgreSQL and Azure Blob Storage for application data and recorded artifacts.

Technologies: Python, Django, Next.js, React, PostgreSQL, LLM APIs, sentence-transformers, Agora SDK, Azure Blob Storage

SecureEval – End-to-End Encrypted File Exchange [github](#)

May 2026

Backend Development | REST APIs | Application Security

- Built a zero-trust file-sharing platform using Next.js, Django REST Framework, and PostgreSQL, with browser-side AES-256-GCM encryption so the server stores only ciphertext.
- Implemented per-recipient RSA-OAEP key wrapping, role-based access control, JWT authentication with rotating refresh tokens, and a tamper-evident SHA-256 hash-chained audit log.

Technologies: Next.js, TypeScript, Django REST Framework, PostgreSQL, Docker, Web Crypto API

SmartCarts – Cloud-Native Microservices E-Commerce [github](#)

Jan 2025

React | Node.js & Express | API Architecture

- Architected a four-service system consisting of an API gateway, user, product, and order services with isolated MongoDB databases and HTTP-based inter-service communication.
- Implemented Redis pub/sub for asynchronous stock updates, containerized the services with Docker, and deployed the platform to Azure Container Apps.

Technologies: React, Node.js, Express, Redis, MongoDB Atlas, Docker, Azure Container Apps

VisionGate – Face Recognition Attendance System [github](#)

Jun 2026

Full-Stack Web Development | FastAPI | Automated Testing

- Built a face-recognition pipeline combining MediaPipe detection, dlib 128-dimensional embeddings, and LBPH classification, validated using 5-fold cross-validation on the LFW benchmark.
- Implemented blink-based liveness detection to mitigate photo spoofing, integrated the pipeline with a FastAPI backend and Next.js dashboard, and developed an **18-test automated test suite**.

Technologies: Python, FastAPI, Next.js, OpenCV, MediaPipe, dlib, SQLite, Git

Additional Project: E-Commerce DevOps Pipeline & Infrastructure [github](#) — Feb 2025

Jenkins CI/CD, commit-tagged Docker images, AWS provisioning with Terraform, and drift-aware deployment via Ansible.

Education

University of Ruhuna

Jun 2022 – Sep 2026

BSc Engineering (Hons) in Computer Engineering, Faculty of Engineering

GPA: 3.46/4.00

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Design, Software Engineering, Database Systems, Cloud Computing